QUESTION 1: WHAT IS ALERT TRIAGE?

DG Interns Hub Cybersecurity Internship – Week 5 Submission

Student Name: Gunjan **Track:** SOC Analyst Tier 1 **Topic:** Alert Triage Fundamentals & L1 Workflow

1. DEFINITION OF ALERT TRIAGE

In a Security Operations Center (SOC), **Alert Triage** is the foundational process of receiving, reviewing, and prioritizing incoming security alerts generated by security tools such as SIEM (Security Information and Event Management), EDR (Endpoint Detection and Response), and IDS/IPS (Intrusion Detection/Prevention Systems).

The term *triage* comes from emergency medicine, where doctors quickly inspect arriving patients to decide who needs immediate life-saving care versus who can wait. In cybersecurity, alert triage works exactly the same way: an L1 SOC analyst rapidly evaluates security events to separate real security incidents from harmless network noise and false alarms.

2. IMPORTANCE OF ALERT TRIAGE IN A SOC

Modern enterprise networks generate thousands of security notifications every single day. Without a structured triage process, security operations would quickly fail due to the following operational reasons:

Preventing Alert Fatigue: Receiving thousands of notifications daily causes analysts to become overwhelmed and exhausted. Triage creates order so analysts don't burn out. **Minimizing Mean Time to Detect (MTTD):** By instantly sorting alerts, high-risk security threats are prioritized, allowing the team to catch active attacks before damage spreads. **SLA Compliance:** SOC teams operate under strict Service Level Agreements (SLAs). Triage ensures critical tickets are reviewed within mandated timelines (e.g., within 15 minutes). **Resource Optimization:** Senior Tier 2 and Tier 3 analysts have limited time. Triage ensures they only handle genuine, high-severity incidents escalated by Tier 1.

3. COMPLETE TIER 1 (L1) SOC ANALYST WORKFLOW

The complete life cycle of an alert, from the moment it is triggered to its final disposition, follows a structured 5-step workflow:

1. **Alert Ingestion & Pickup:** The SIEM detects abnormal activity and triggers an alert in the queue. The L1 analyst assigns the ticket to themselves.
2. **Initial Context Gathering:** The analyst opens the raw log data to collect critical context: source/destination IP addresses, hostnames, timestamps, user accounts, and triggering rule logic.
3. **Investigation & Enrichment:** Indicators (IPs, hashes, domain names) are queried in Threat Intelligence platforms (VirusTotal, AbuseIPDB) and cross-checked with internal Active Directory context.
4. **Classification (Decision Point):** The analyst categorizes the event as a **True Positive** (real security incident), **False Positive** (harmless misconfiguration), or **True Negative**.

5 **Closure or Escalation:** If benign, document findings and close ticket. If a critical True Positive, apply standard L1 response playbooks and escalate to Tier 2 with detailed notes.

4. PRACTICAL EXAMPLE OF AN ALERT INVESTIGATION

Scenario: High Volume of SSH Failed Logins

Triggered Alert: "Multiple SSH Authentication Failures" from source IP `192.168.1.105` targeting internal host `SERVER-DB-01`.

L1 Analyst Investigation Process:

Log Inspection: The analyst checks SIEM logs and finds 60 failed login attempts targeting the `root` account within 90 seconds. **Asset Context Check:** `SERVER-DB-01` is a critical production database server holding customer records. **Source Identification:** IP `192.168.1.105` belongs to a developer workstation on the internal subnet. **Verification:** The analyst contacts the assigned developer via internal chat. The developer confirms they mistyped their password while running an automated deployment script. **Conclusion:** Classified as a **False Positive** (misconfigured administrative script). The analyst documents the ticket with communication proof and closes it.

5. KEY TERMINOLOGY & LEARNING REFERENCES

SIEM: Security Information and Event Management system for log aggregation and correlation. **Alert Fatigue:** Operational burnout caused by high volumes of repetitive alerts. **MTTD:** Mean Time to Detect — key metric measuring how fast a threat is identified.

References:

• MITRE ATT&CK SOC Operations Guide: **https://attack.mitre.org/**
• CISA Security Operations Center Best Practices: **https://www.cisa.gov/**